

SEVERNOYE, Russian Federation

WMO: 294180

Lat: 56.3377N

Lon: 78.3575E

Elev: 127

StdP: 99.81

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-36.5	-33.2	-39.2	0.1	-36.4	-36.3	0.1	-32.9	10.0	-11.4	8.9	-7.3	0.6	350	0.630

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	28.5	18.9	26.9	18.4	25.4	17.6	20.6	26.0	19.7	24.8	18.7	23.4	3.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.7	13.8	23.2	17.8	13.0	21.9	16.9	12.2	21.2	60.3	25.9	57.0	24.8	53.9	23.4	24.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.4	7.9	6.9	-40.5	30.9	4.0	1.5	-43.4	31.9	-45.7	32.8	-48.0	33.7	-50.9	34.8		
			-40.5	22.4	3.8	0.9	-43.2	23.1	-45.4	23.6	-47.5	24.1	-50.3	24.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	0.9	-18.9	-16.0	-6.8	2.5	10.7	16.3	18.2	15.2	9.3	2.3	-7.8	-15.3
	DBStd	14.22	8.23	7.36	5.96	5.89	5.41	4.31	3.34	3.65	4.35	4.99	7.92	8.53
	HDD10.0	4070	897	728	522	232	59	6	0	3	64	242	534	783
	HDD18.3	6446	1156	961	780	474	241	87	44	108	272	497	784	1041
	CDD10.0	748	0	0	0	8	82	193	253	165	43	4	0	0
	CDD18.3	83	0	0	0	0	5	25	39	12	1	0	0	0
	CDH23.3	845	0	0	0	1	98	263	336	133	14	0	0	0
	CDH26.7	158	0	0	0	0	18	53	65	20	1	0	0	0
Wind	WSAvg	2.8	2.8	3.0	3.4	3.3	3.1	2.3	2.0	2.1	2.5	3.0	3.1	3.3
Precipitation	PrecAvg	476	23	18	19	25	37	63	79	69	44	39	35	28
	PrecMax	655	45	35	52	68	85	150	225	131	97	75	69	55
	PrecMin	317	3	1	1	7	9	10	11	13	13	16	9	7
	PrecStd	97	13	7	12	15	19	35	46	31	24	18	13	10
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.3	0.9	7.3	21.7	29.1	30.2	30.6	28.9	25.1	18.3	6.7	1.2
		MCWB	-1.1	-0.4	3.0	12.5	16.0	19.2	20.9	19.6	15.9	10.9	5.0	0.3
	2%	DB	-2.6	-1.0	4.6	17.3	26.0	28.2	28.4	26.5	22.1	15.1	4.7	-0.1
		MCWB	-3.3	-1.8	1.3	10.1	15.0	18.5	19.5	18.7	14.4	9.3	3.5	-0.8
	5%	DB	-4.4	-2.8	2.8	14.3	23.6	26.4	26.9	24.7	19.2	11.9	2.7	-1.8
		MCWB	-5.1	-3.6	0.5	8.3	13.8	18.1	19.0	17.8	13.1	7.5	1.5	-2.5
	10%	DB	-6.7	-5.0	1.4	11.2	20.8	24.5	25.2	22.6	16.8	9.4	1.0	-3.9
		MCWB	-7.3	-5.8	-0.4	6.5	12.7	17.1	18.5	16.9	11.8	6.4	0.1	-4.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.9	-0.1	3.2	13.1	17.6	21.7	22.2	21.1	17.2	11.8	5.4	0.6
		MCDB	-0.3	0.7	6.6	20.0	27.1	27.2	28.1	26.2	23.0	16.8	6.0	0.9
	2%	WB	-3.3	-1.9	1.8	10.7	16.0	20.0	21.0	19.7	15.2	9.8	3.5	-0.8
		MCDB	-2.7	-1.0	3.8	15.8	24.1	25.8	26.3	25.3	20.2	14.4	4.6	-0.1
	5%	WB	-5.1	-3.6	0.8	8.7	14.5	19.0	20.1	18.7	13.8	8.1	1.7	-2.5
		MCDB	-4.5	-2.9	2.5	13.6	21.9	24.7	24.9	23.0	18.3	11.2	2.5	-1.9
	10%	WB	-7.3	-5.8	-0.3	6.9	13.1	17.9	19.2	17.5	12.4	6.6	0.3	-4.5
		MCDB	-6.7	-5.0	1.4	11.0	19.9	23.1	23.6	21.6	16.2	9.2	0.9	-3.9
Mean Daily Temperature Range	5% DB	MDBR	8.4	9.8	9.7	9.4	11.3	10.8	10.2	10.3	9.6	7.2	6.3	7.8
		MCDBR	8.1	7.5	9.3	13.6	15.4	13.9	12.8	14.0	14.0	12.2	6.0	7.2
		MCWBR	7.9	7.1	7.0	7.8	6.8	6.0	5.5	7.0	7.4	7.3	5.4	7.1
	5% WB	MCDBR	8.0	7.5	7.7	12.7	14.2	12.1	11.1	11.9	12.4	11.2	5.7	7.3
		MCWBR	7.9	7.3	6.2	7.7	7.1	5.9	5.4	6.6	7.1	7.2	5.3	7.2
Clear-Sky Solar Irradiance	taub	0.280	0.313	0.368	0.406	0.402	0.404	0.434	0.412	0.364	0.349	0.310	0.283	
	taud	2.023	2.004	1.927	2.069	2.235	2.301	2.213	2.252	2.347	2.234	2.077	1.966	
	Ebn at Noon	575	711	760	801	836	838	801	793	784	676	529	447	
	Edh at Noon	64	99	139	141	127	121	129	117	92	79	61	53	
All-Sky Solar Radiation	RadAvg	0.82	1.85	3.39	4.60	5.22	5.83	5.27	4.17	2.88	1.43	0.68	0.51	
	RadStd	0.07	0.11	0.18	0.38	0.47	0.60	0.36	0.41	0.30	0.14	0.13	0.07	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page